# VAC TRUCKS

## IN CIVIL CONSTRUCTION

Foreman field guide: planning, daylighting, documentation and cost control

PAGES 1-3

### YOU NEED TO KNOW SECTION

The short version foremen need before booking or directing the truck.

PAGES 4-13

### SUPPLEMENTAL TEACHING + GOOD PRACTICES

Use these pages for training, coaching and cleanup after the toolbox talk.

WHY THIS MATTERS

**A vac truck can protect live utilities, prove field conditions and keep a crew moving.**

But without a clear request, it becomes a four-hour minimum invoice with headlights.

FIELD TRAINING

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YOU NEED TO KNOW

# The field rules before the truck rolls

The short version: use the vac truck for safety, information and production. Everything else is expensive noise.

1

## Use it where it adds value

Live utilities, crossings, tight areas, unknown conflicts, tie-ins and daylight holes ahead of excavation.

2

## Book a defined scope

Exact location, number of holes, expected depth, utility being exposed and expected duration.

3

## Be ready before arrival

Current locates, visible paint and flags, access, traffic control, water source and disposal plan.

4

## Daylight ahead where practical

Do the exposure before the excavator reaches the conflict, not while the crew stands around waiting.

5

## Record the exposure

Photo, depth to top, utility type, offset or station, and a physical marker where practical.

6

## Keep backup holes ready

If the main hole is delayed or finished early, move to useful secondary holes instead of idling the truck.

**Stop and reassess if the locates, drawings or exposed conditions do not agree.**

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YOU NEED TO KNOW

# Booking and cost control

A vac request needs scope, duration and the reason why. That is how you avoid waste and catch billable work.

| 4 h | 2 h | 0 | WHY |
|---|---|---|---|
| TYPICAL MINIMUM | OFF-SITE DUMP EVENT | IDLE HOLES ACCEPTABLE | BEFORE YOU BOOK |

## Minimum request before booking

Where is it?
How many holes?
What utility or conflict?
Expected depth and size?
Expected duration?
Are locates, access and traffic control ready?
What are the backup holes?

## Ask the WHY before treating it as cost

Changed ground condition?
Utility not shown or not matching locates?
Owner or inspector direction?
Missing information from drawings?
Design conflict discovered in the field?

That review can turn crew time and vac time into a T&M event instead of pure overhead.

**Backup holes are not optional planning fluff. They are how you keep a 4-hour minimum productive when the first task hits a delay.**

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SUPPLEMENTAL

# What a vac truck is actually for

Non-destructive excavation is valuable because it lowers utility strike risk and gives the crew better information before mechanical excavation.

### Positive exposure

Daylight known or suspected utilities so the crew can see the actual line, depth, material and alignment before equipment gets close.

### Tight or congested areas

Use vacuum excavation where mechanical digging has a poor risk profile: crossings, service clusters, tie-ins and conflict points.

### Targeted excavation

Potholes, test holes, limited cleanout and exposure work. The key word is targeted. A vac truck is not a substitute for an excavator.

THE FIELD STANDARD

**Use the vac truck where it adds safety or information. Once it has done that job, move it on.**

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SUPPLEMENTAL

# Plan the vac before it arrives

The truck should arrive to execute a plan, not to help everyone discover that nobody had one. Humanity survives another day.

### 1. Scope

Exact location
Number of holes
Approx. depth and size
Utility or conflict being exposed

### 2. Readiness

Current locates
Paint and flags visible
Access confirmed
Traffic control ready

### 3. Production

Expected duration
Crew ready to use results
Excavation sequence planned
Backup holes identified

### 4. Logistics

Water source
Spoil / slurry disposal
Hose reach
Truck staging and exit route

**If the foreman cannot describe the scope and expected duration, the request is not ready for dispatch.**

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SUPPLEMENTAL

# Daylight ahead of the excavator

Where practical, expose the critical utilities before the main excavation crew arrives at the conflict.




**LOCATES**

Verify current docs

**VAC**

Expose the conflict

**RECORD**

Depth + photo + offset

**MARK**

Preserve the information

**EXCAVATE**

Mechanical excavation

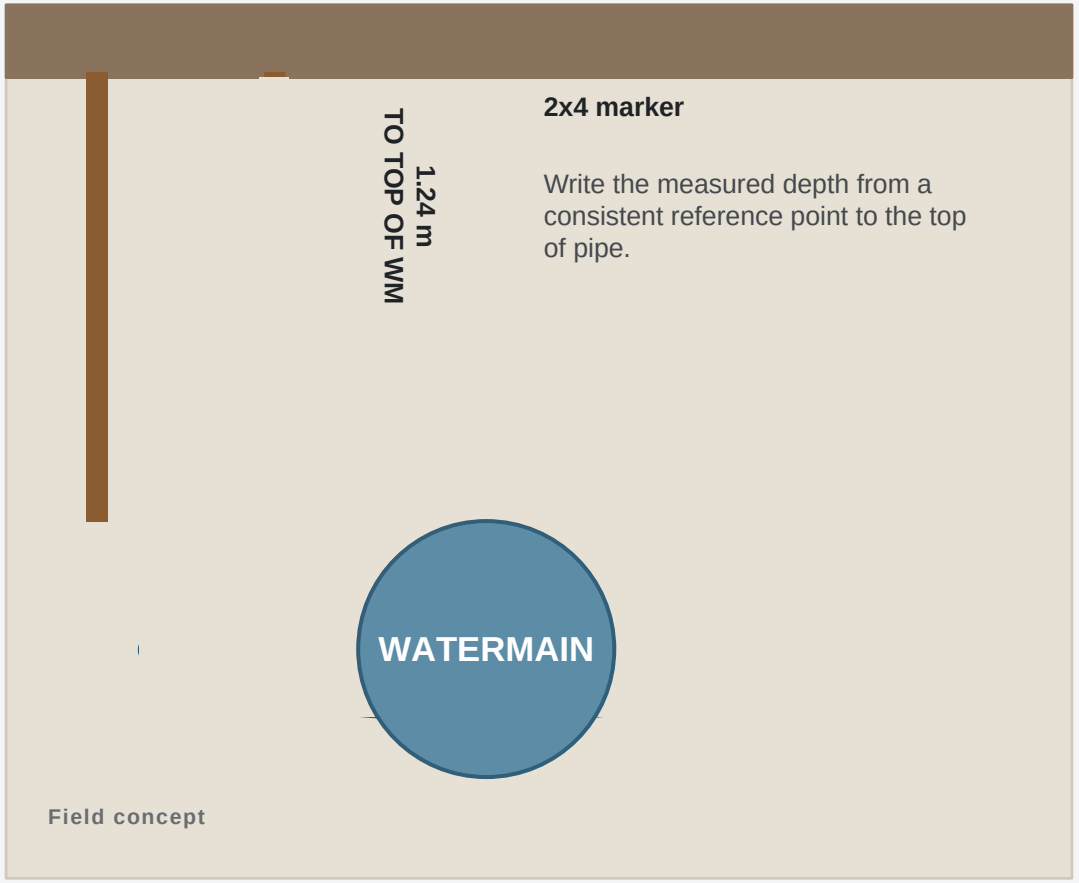
**Bad sequence: excavator reaches the conflict, stops, then everyone waits for a vac truck that was never planned.**

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SUPPLEMENTAL

# Do not daylight the same utility twice

A daylight hole has value only if the information survives after the vac truck leaves.




## Minimum record

- ✓ Photo showing the exposed utility and surrounding context
- ✓ Utility type, material and diameter if visible / known
- ✓ Measured depth to top of utility from a clear reference
- ✓ Offset or station reference so the hole can be found again
- ✓ Physical marker preserved where practical: stake, 2x4, paint
- ✓ Information communicated to the operator / pipe layer before excavation

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SUPPLEMENTAL

# A simple daylight log beats memory

Keep it boring, consistent and easy enough that people actually use it.

| | |
|---|---|
| **Date / Hole ID** | 2026-09-20 / DH-07 |
| **Location** | Sta. 12+460, east curb lane |
| **Utility** | 150 mm watermain |
| **Depth to top** | 1.24 m below reference |
| **Offset** | 0.85 m north of paint mark |
| **Condition** | Good, no visible damage |
| **Photo** | IMG_4472 / IMG_4473 |
| **Marked** | 2x4 + blue paint |
| **Crew note** | Excavate from south, expose by hand near crossing |

## Why it matters

The pipe layer should not be rediscovering the same line three days later with a shovel while everyone argues about what somebody remembers.

## Keep one source of truth

Use the company daily report, a shared log, marked-up drawing, or a simple spreadsheet. The format matters less than consistency.

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SUPPLEMENTAL

# Safety rules do not disappear because the tool is a hydrovac

Vacuum excavation reduces some risks. It introduces others, and utility-owner requirements still control the work.

- ✓ Confirm utilities in and near the excavation are located and marked before excavation begins.
- ✓ Review the utility owner requirements, including any pressure limits or special procedures.
- ✓ Keep workers clear of the vacuum tube and control splash, noise, traffic and slippery surfaces.
- ✓ Use manufacturer instructions, trained operators and functioning dead-man / shutdown controls.
- ✓ Stop and reassess if the field markings, drawings or exposed conditions do not agree.
- ✓ Support exposed services where required and protect open daylight holes after the work.

ONTARIO

**O. Reg. 213/91, s. 228 requires gas, electrical and other services in and near an excavation to be located and marked. If a hazardous service cannot be shut off or disconnected, the service owner is to be requested to supervise its uncovering.**

Company procedures and utility-owner requirements can be more restrictive. Follow the most applicable requirement for the work.

*Sources: Ontario O. Reg. 213/91 s. 228; IHSA Safe Practice Guide SPG4; utility-owner requirements.*

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SUPPLEMENTAL

# The six fastest ways to waste a vac truck

Most lost time is planning failure wearing a hard hat. Nature is healing, apparently.

### No exact scope

Just start daylighting around here.

### No access

Cars, bins, barriers or equipment block the work area.

### No traffic setup

Truck arrives before the lane closure or spotters are ready.

### No disposal plan

Tank is full and nobody knows where it can legally dump.

### Crew not ready

The truck exposes the utility, then waits while the real work catches up.

### One-hole thinking

No backup holes are ready when the first exposure is complete.

**Keep a short backup list of useful daylight holes. If the main task is delayed, the truck can stay productive.**

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SUPPLEMENTAL

# Cost control is about truck-hours and recoverability

Do not just ask whether a vac truck is needed. Ask why it is needed and who should carry the cost.

| 4 h | 2 h | 3+ | T&M |
|---|---|---|---|
| MINIMUM CALL-OUT | DUMP OFF SITE | BACKUP HOLES READY | POSSIBLE RECOVERY |

### Plan around the minimums

Assume the truck is a 4-hour event. If it has to leave site to dump, assume roughly another 2 hours of impact before it is productive again. Build a backup list so the minimum gets used on meaningful holes.

### Turn surprises into records

When the reason for the vac is changed ground, a utility not shown on locates, an unexpected conflict, or owner / inspector direction, capture it. Photos, notes, times, location and instruction chain can support a T&M event.

**The same truck can be pure cost or billable production. The difference is usually the foreman understanding the WHY and documenting it.**

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SUPPLEMENTAL

# Preserve the work for the excavation crew

The vac truck creates information. The site has to keep that information useful.

1. **Measure** — Depth to top of utility from a repeatable reference.
2. **Photograph** — Wide context photo plus a clear view of the exposed service.
3. **Mark** — 2x4, stake or paint. Write the depth directly on the marker where practical.
4. **Log** — Hole ID, utility, station / offset, depth and notes in one shared record.
5. **Communicate** — Brief the operator, pipe layer and foreman before mechanical excavation reaches it.

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SUPPLEMENTAL

# References and use notes

Use this deck as a field-training aid, not as a replacement for project-specific procedures or utility-owner requirements.

### Ontario Regulation 213/91 - Construction Projects

Section 228: precautions concerning underground services, including locating and marking services before excavation.

### IHSA SPG4 - Excavating with Vacuum Technology in the Vicinity of Electrical Utilities

Safe practice guidance for hydrovac operation, work zones, vacuum hazards, equipment controls and utility exposure.

### Ontario One Call

Locate request rules and validity requirements. Confirm the current locate status before excavation.

### Utility-owner requirements

Examples include gas and electrical utilities. Requirements can include tolerance zones, pressure limits, supervision and support of exposed services.

**Training use: first 3 pages for the field talk, remaining pages for coaching and follow-up.**

**Sources checked September 20, 2026**

ontario.ca/laws/regulation/910213 | ihsa.ca | ontarioonecall.ca | utility-owner requirements

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